n-types are closed under dependent sums

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:ntypes-sigma' (source lines 140-143, Theorem 7.1.8); the proof environment starts after one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own lead-in sentence announcing the closure results of the section.

Editorial note. The retained counter values (chapter 7, section 1, theorem offset 7) make the retained theorem print as Theorem 7.1.8, matching the original book numbering.

Editorial note. Both external references point at results that the retained proof invokes by label, one of them in the same file; the counted proof is complete without them.

Editorial scope. One original result of the HoTT Book, counted once: closure of n-types under dependent sums (Theorem 7.1.8).

Allowed standard prerequisites (editorial).

- n-types and their preservation under equivalence. If X is equivalent to an n-type then X is an n-type.. It is applied at source line 159 to replace the path space of the sigma type by an equivalent sigma type.
- Paths in a sigma type. (a_1,b_1) = (a_2,b_2) is equivalent to Sigma p : a_1 = a_2, trans p b_1 = b_2.. It produces the equivalence displayed at source line 158 that the induction step works with.
- The induction hypothesis on n-types. Contractible types and their path spaces behave as in the definition of the hierarchy.. The base case at lines 148-151 and the step at lines 153-161 are both reduced to the inductive hypothesis.

Source reference. Original theorem 2.7.2: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1853–1853. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original corollary 7.1.5: sources/hott-hlevels/pinned-original-source.tex, source lines 96–96. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

We now show that n-types are preserved by most of the type forming operations.

Theorem 7.1.8. *Let $n \geq -2$, and let $A : \mathcal{U}$ and $B : A \to \mathcal{U}$. If A is an n-type and for all $a : A$, $B(a)$ is an n-type, then so is $\sum_{(x:A)} B(x)$.*

Proof. We proceed by induction on n.

For $n = -2$, we choose the center of contraction for $\sum_{(x:A)} B(x)$ to be the pair (a_0, b_0), where $a_0 : A$ is the center of contraction of A and $b_0 : B(a_0)$ is the center of contraction of $B(a_0)$. Given any other element (a, b) of $\sum_{(x:A)} B(x)$, we provide a path $(a, b) = (a_0, b_0)$ by contractibility of A and $B(a_0)$, respectively.

For the inductive step, suppose that A is an $(n+1)$-type and for any $a : A$, $B(a)$ is an $(n+1)$-type. We show that $\sum_{(x:A)} B(x)$ is an $(n+1)$-type: fix (a_1, b_1) and (a_2, b_2) in $\sum_{(x:A)} B(x)$, we show that $(a_1, b_1) = (a_2, b_2)$ is an n-type. By original theorem 2.7.2 we have

$$((a_1, b_1) = (a_2, b_2)) \simeq \sum_{p:a_1=a_2} (p_*(b_1) =_{B(a_2)} b_2)$$

and by preservation of n-types under equivalences (original corollary 7.1.5) it suffices to prove that the latter is an n-type. This follows from the inductive hypothesis. □